

An Automated Platform for GNSS Processing and Time-Series Analysis in a Complex Geodynamic Region

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Presentation Structure

- GNSS Networks in Greece
- Upgrade Processing and analysis platform
- Results & Outputs
- Time series analysis results
- Discussion / Conclusions



Motivation

Routine GNSS processing and site/network monitoring is crucial, because:

- Greece lies in a region of utmost tectonic and volcanic unrest (e.g. active volcano in Santorini isl.),
- results & products are important to a series of fields spanning the whole range of Geosciences,
- helps us follow and apply state-of-the-art technologies in GNSS analysis & Satellite Geodesy and expand & modernize our research activity,
- contribute to the GNSS/EUREF community and be involved in ongoing/future projects,
- develop new features for high rate processing.
- improve our academic services (NTUA is a University)

Throughout the last years, routine processing & monitoring has helped us gain a more thorough view of the complex tectonic and volcanic setting of Greece.

The Data Set

Routine processing for precise positioning, assumes a well established, credible dataset (metadata).

Currently we process whatever we can get our hands on ...

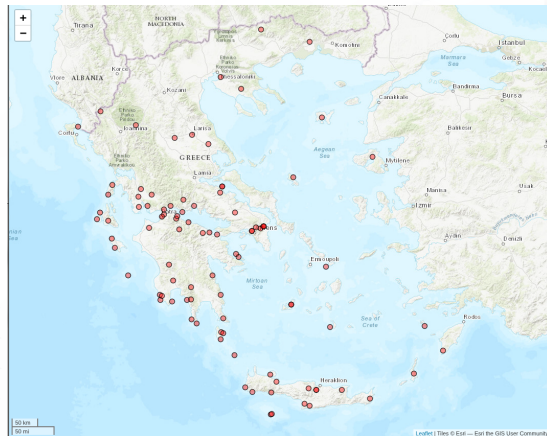
Problems:

- Inhomogenous dataset (RINEX of various versions, raw files, etc).
- Various maintainers, different mentalities.
- Different acquisition methods/rates.
- No log files for maintainers with no geodetic interest (e.g. surveying companies).
- Wide variety of equipment (not always included in `atx` files).

Network GREECE

Network Greece includes the majority of the available sites (≈ 100) but not all of them are (always/currently) active. Various providers but all with geodetic interest & equipment.

- covers all of Greece
- different (geodetic type) equipment
- credible time-span (early 2004 - now)
- all free available GNSS data
- large data gaps & inactive stations

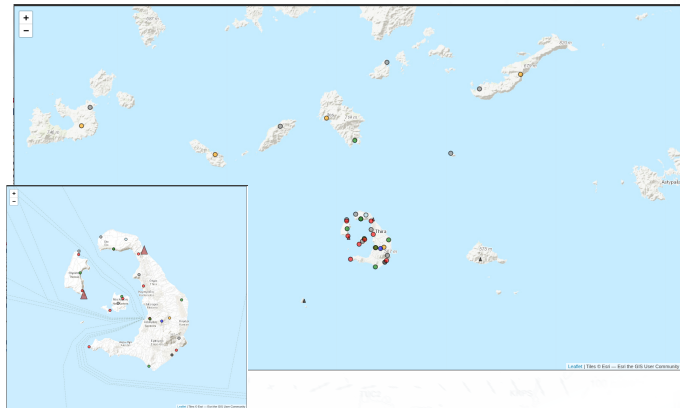


-> Network GREECE

Network Santorini

The Santorini Volcano ...

- credible time-span
- only covers the Corinth Rift
- different providers (including surveying & cadastral services)
- no log files & equipment changes



-» Network Santorini

Network HEPOS

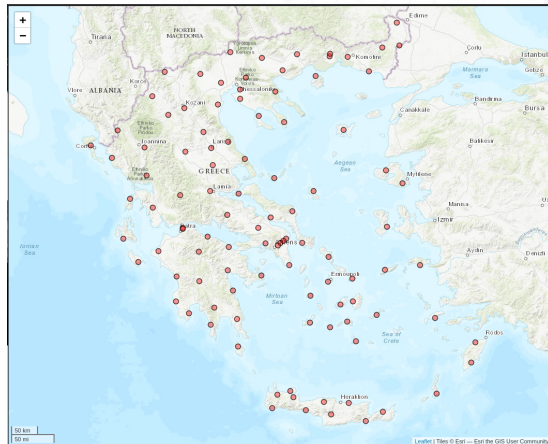
Network HEPOS

- covers all of Greece
- homogeneous equipment
- credible time-span (mid 2007 - now)
- only four years available for processing up to now

*Data availability and
funding from
Hellenic Cadastre*



HELLENIC CADASTRE



-» Network HEPOS

GNSS data analysis at DSO

GNSS data analysis

- Bernese v5.4 ([Dach, 2015](#))
- PRIDE-PPPAR ([Geng et al., 2019](#))
- GipsyX ([Bertiger et al., 2020](#))

Time series analysis

- Hector v2.1 ([Bos et al., 2012](#))
- Python/C++ routines

Solutions

- Daily solution
 - Rapid solution - 1 day after
 - Final solution - 12 days after
- High-rate analysis (>1 Hz)
 - data from network HEPOS - 1 Hz



The Scheme

The core tool/software is **Bernese GNSS Software v5.4**.

Integration with

- **PostgreSQL** database,
- **Python** module (product/data downloading, pre-processing, driving cron jobs, etc)
<https://github.com/DSOlab/autobern>
- **Time-series** analysis (integrated in routine processing on regular intervals)
- **Strain Rates** via StrainTool (on user demand)

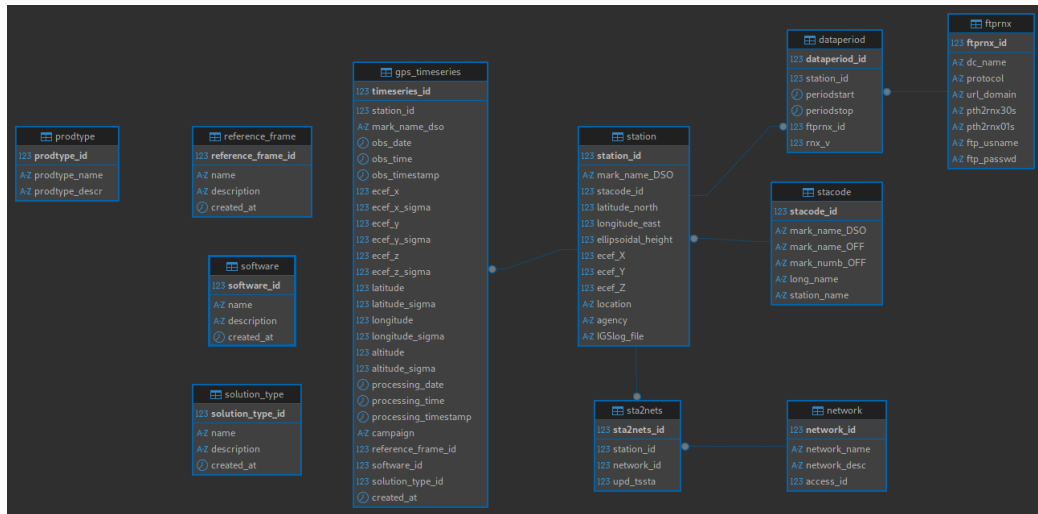


Switch to IGS20 - REPRO2026

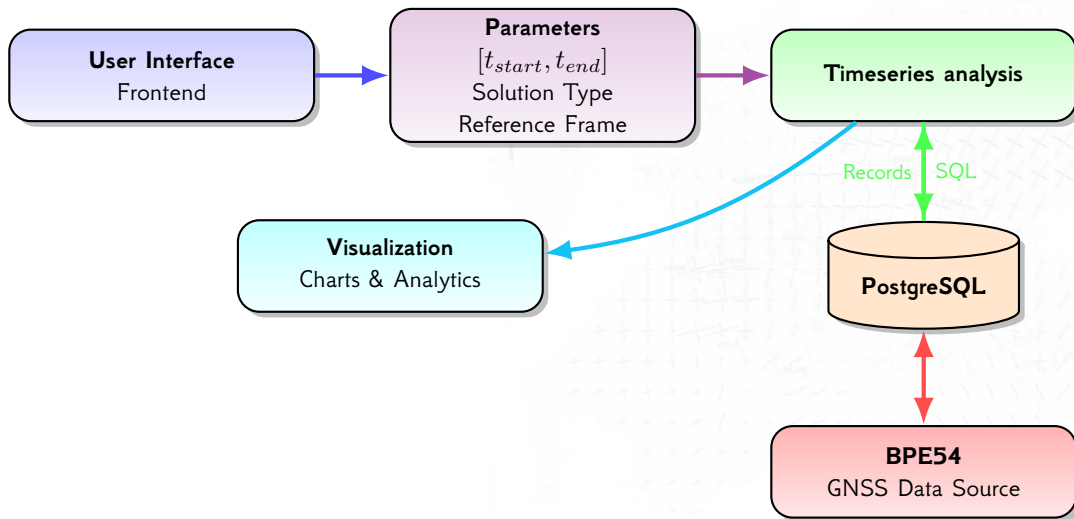
- Reference frame [IGS20](#) > [IGb20](#) > [IGc20](#)
- Oceans loading corrections ([NTUA_FES2014b](#))
- [VMF3](#) for tropospheric modeling
- ATX file: [igs20.atx](#)
- Long filename of products according to new IGS convention

	Server	Start	Finish	REVIEW	UPLOAD EPN	start day	proc days	end day	Last script run	COMMENTS
1995										
1996										
***	***	***	***	***	***	***	***	***	***	***
2018										week=2008 start BIA, jul 1st/182
2019										
2020	bernese1	13.06.2026		ongoing		001	224	366	Wed Jun 17 21:34:48 UTC 2026	
2021	paraskevas	12.06.2026		ongoing		001	265	365	Wed 17 Jun 08:29:09 EEST 2026	
2022	dso01	12.06.2026		ongoing		001	262	365	Wed Jun 17 08:32:43 EEST 2026	2022-331 -> change to new long
2023	bernese1	05.06.2026	12.06.2026	4REVIEW	OK ALL	001		365	FINISHED	
2024	paraskevas	02.06.2026	11.06.2026	4REVIEW	OK ALL	001		366	FINISHED	
2025A	dso01	02.06.2026	04.06.2026	4REVIEW	OK ALL	001		050	FINISHED	
2025B	dso01	06.06.2026	11.06.2026	4REVIEW	OK ALL	051		365	FINISHED	switch to IGB20 02.02.2025
2026	dso01					033		050		USE IGS20 check the results
2026B	vincenty	16.06.2026		4REVIEW		001		010	FINISHED	
2026C	vincenty	16.06.2026		ongoing		011	075	147	Tue Jun 16 11:54:16 EEST 2026	switch to IGC20 11.01.2026
2027										

Turn to new DB



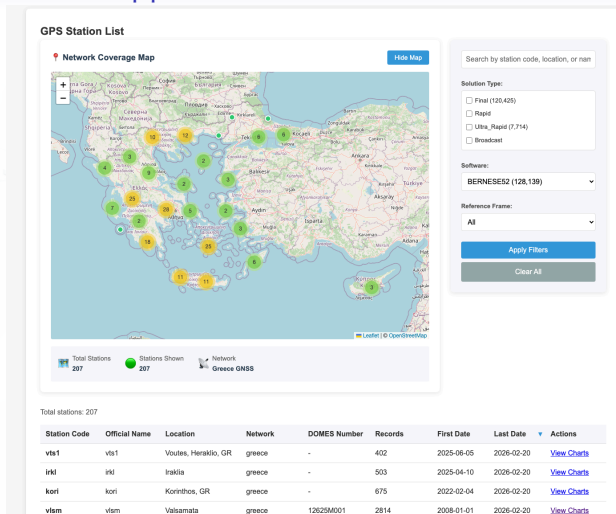
Platform flowchart



Presentation Layer - Main Application Screen

- User entry point to the application.
- Selection of basic parameters:
 - ◇ GNSS station
 - ◇ Solution type
 - ◇ Reference frame
- Filtering and reordering capability

Live app (beta version):



<http://83.212.80.177:8585/>

Results & Output

4. Solution Indetifiers

Array of Objects

expand

5. PCF Variables

Array of Objects

expand

6. Saved products

Array of Objects

expand

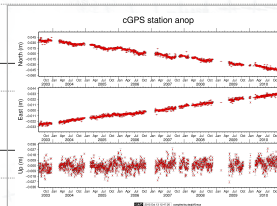
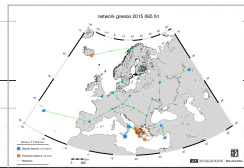
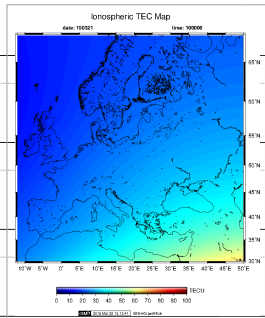
7. Warnings

Array of Objects

expand

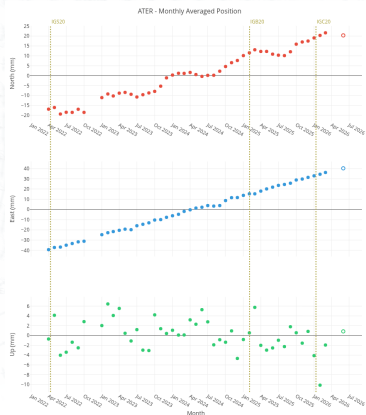
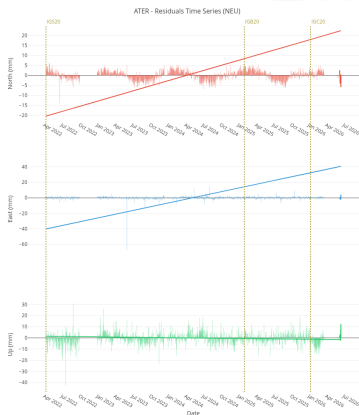
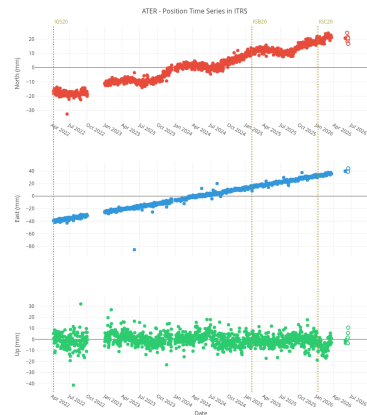
8. Ambiguity Resolution Summary

Array of Objects



Baseline	sta1	sta2	length (km)	Method	N. of Amb.	Percentage	Satellite system
AUKL	AUT1	KLOK	139.7	pbnl	74	54.1	GPS
AULE	AUT1	LEMN	199.6	pbnl	60	55	GPS
KCTL	KATC	TILO	59	pbnl	50	90	GPS
KLRL	KLOK	RLSO	174.2	pbnl	74	41.9	GPS

Online results



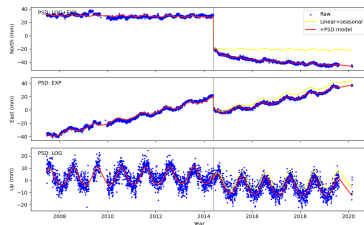
Time series analysis

Linear+Seasonal+PSD

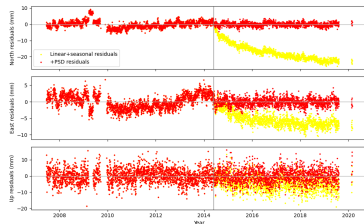
$$y(\delta t) = A\delta t + B + C_1 \sin(\omega_1 \delta t) + D_1 \cos(\omega_1 \delta t) + C_2 \sin(\omega_2 \delta t) + D_2 \cos(\omega_2 \delta t) \\ + \sum_{k=1}^n J_k H(t - t_k) + \sum_{i=1}^n A'_i \log\left(1 + \frac{\delta t}{T'_i}\right) + \sum_{j=1}^n A_j \left(1 - e^{-\frac{\delta t}{\tau_j}}\right)$$

Preliminary Results in Greece

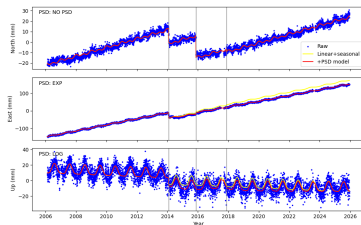
LEMN.txyz2 trajectory



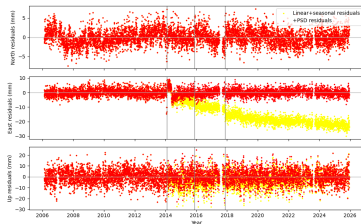
LEMN.txyz2 residuals



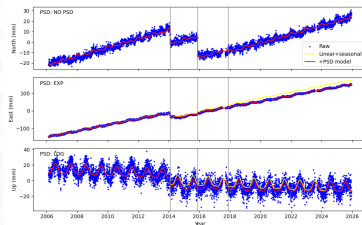
VLSM.txyz2 trajectory



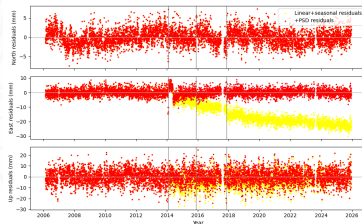
VLSM.txyz2 residuals



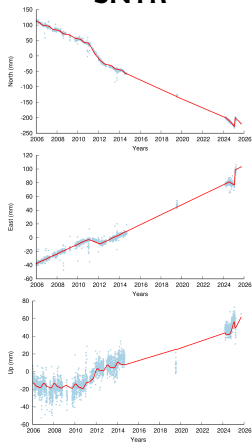
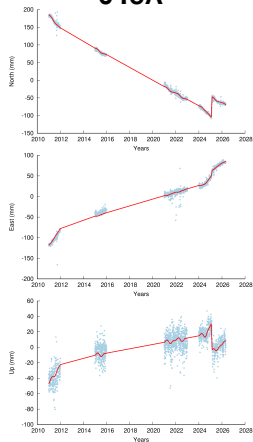
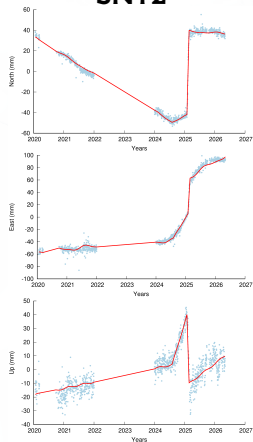
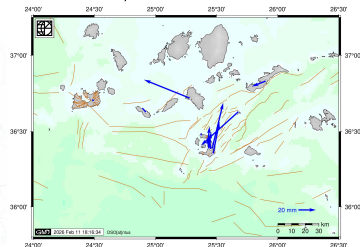
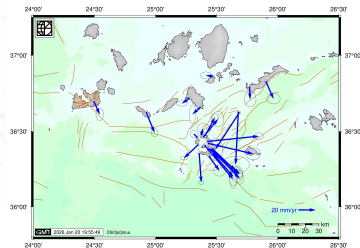
VLSM.txyz2 trajectory



VLSM.txyz2 residuals



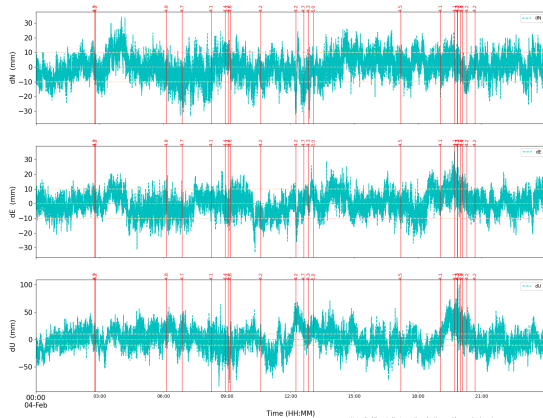
Specific case - Santorini

SNTR**048A****SNT2****Horizontal displacements 20250201 - 20250228 [IGS20]****Horizontal velocities 20250303 - 20250808 [IGS20]**

PPP processing for High-rate data

effect Santorini Volcano crisis (05.02.2025)

048A_2025_035



*Note: Red lines indicate earthquake times with magnitude > 4.

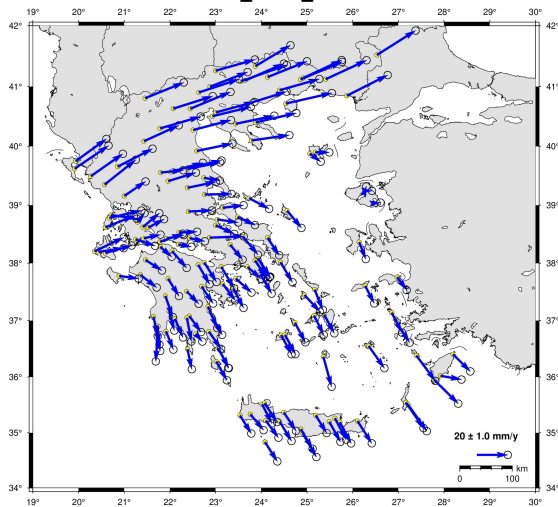
048A_2025_036



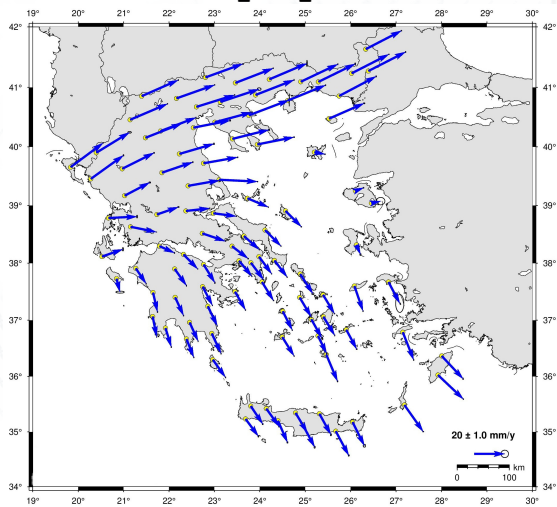
*Note: Red lines indicate earthquake times with magnitude > 4.

REPRO2026: Compare velocity field of different networks

DSO_GRC_2026



DSO_HPS_2026



Discussion / Conclusions

- Greece is located in a complex tectonic background with many changes in the kinematics of the area.
- Routine processing and monitoring are very important and revealing for Greece; products are requested by and disseminated to a wide range of Geoscientists.
- A dense velocity field for accurate estimation of ground/tectonic motions in the region will help to develop a stable local reference frame and the connection of the region with the global and European reference systems.
- The continuous monitoring of the networks gives useful results for the effect of strong earthquakes or other “abrupt” phenomena; small, dense networks are of great help (even with instrumentation of non-geodetic accuracy).
- DSO is upgrading its scientific output and processing activity via new datasets and real time processing.
- Our plan: grow up our team and be more involved to the GNSS/EUREF community.

Thank you for your attention! 🇸🇩🇬🇪



References I



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Geng, Jianghui, Xingyu Chen, Yuanxin Pan, Shuyin Mao, Chenghong Li, Jinning Zhou, and Kunlun Zhang (Oct. 2019). "PRIDE PPP-AR: an open-source software for GPS PPP ambiguity resolution". en. In: *GPS Solutions* 23.4, p. 91. issn: 1080-5370, 1521-1886. doi: [10.1007/s10291-019-0888-1](https://doi.org/10.1007/s10291-019-0888-1). URL: <https://link.springer.com/10.1007/s10291-019-0888-1> (visited on 06/18/2026) (cit. on p. 8).